Customer Support Chatbot MLOps and Agentic AI System

Enterprise MLOps & Agentic AI System

Active Model: `v1.0.1`

Framework: `Scikit-Learn + TF-IDF`

Orchestrator: `5-Agent Pipeline`

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- Batch Processing & CSV
- MLOps & Model Governance
- Human Review & Retraining

© 2026 Customer Support Chatbot MLOps and Agentic AI System

Real-Time Multi-Agent Complaint Analyzer

Run the complete 5-Agent orchestration pipeline on an individual complaint.

Quick-Load Sample Scenario:

Custom Complaint

Customer Name:

A

Complaint Narrative:

When will ordered things arrive?

Run Multi-Agent Analysis Pipeline

Pipeline Completed in 411.11 ms | Trace ID: `TRC-2DFB037F`

Predicted Category	Urgency	Sentiment	Priority Tier	Target SLA
Account A…	High	Dissatisfied	P2-High	8 Hours
↑ 26.8% Conf	↑ 38.0% Conf	↑ 48.4% Conf	↑ Score: 40/100	↑ Identity & Security …

AI Explainability (Top Driver Tokens)

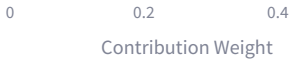


Priority Routing Rationale

Assigned Department: `Identity & Security Desk`

Legal / Regulatory Threat: No

- Base urgency [High]: +35 pts
- Customer dissatisfaction: +5 pts



Identified 1 primary driver term(s): 'arrive' (+0.5024).

📋 Recommended Remediation Playbook (SOP)

- 1. Verify customer identity via two-factor biometric callback or secondary knowledge-based authentication.
- 2. Clear security lockout flag in IAM directory and reset multi-factor device pairing.
- 3. Guide customer through secure portal onboarding and mobile application re-authentication.
- 4. Monitor session for 24 hours to confirm stable account access.

Financial Compensation Advice: No direct monetary compensation required; standard issue remediation.

Statutory Compliance Notice: `Standard internal service level agreement applies.`

✉️ Drafted Customer Communication

```
Dear A,

Thank you for contacting our customer resolutions team. We have received your inquiry and classified it under our P2-High

Action in progress: Priority Tier 2 Callback & Investigation.
Our commitment: A resolution specialist will follow up with you within 8 hours with an update or final determination.

We deeply appreciate your patience and the opportunity to make this right.

Sincerely,
Executive Customer Solutions Team
```

🛠️ View Multi-Agent Execution Trace Waterfall

[ValidationAgent] — Data Sanitation & PII Redaction (0.12 ms • Status: SUCCESS)

[ClassificationAgent] — Multi-Task ML Inference & Explainability (321.93 ms • Status: SUCCESS)

[InvestigationAgent] — Root Cause & Historical Pattern Investigation (58.2 ms • Status: SUCCESS)

[PriorityRoutingAgent] — Priority Scoring & Department Routing (0.07 ms • Status: SUCCESS)

[RecommendationAgent] — Remediation Playbook & Response Synthesis (0.04 ms • Status: SUCCESS)